ME 537 Final Exam - Fall 2023

3 Hours Timed, No Book, Two sheet of notes front and back allowed
Calculators, Python, or MATLAB are allowed (for calculations, plotting functions, and simple matrix math - not for extra notes, existing robotics libraries, or symbolic manipulation)

Do not open the test until you are ready to begin. Complete the exam in one sitting. Please show all work and list ALL assumptions. If you make your work clear and understandable, it will be easier for me to give credit where credit is due. Clearly label your solutions by boxing or underlining them. I have tried to leave sufficient space to work each problem, but you may use your own paper if necessary or if you prefer.

Name: Nathan Ludlow
Starting Time: 12/19/2023 11:28 Pm
Ending Time: 12/20/2023 2:04 Pm

Problem 1: ____ /8
Problem 2: ____ /15
Problem 3: ____ /16
Problem 4: ____ /12
Problem 5: ____ /9
Problem 6: ____ /12
Problem 7: ____ /12
Problem 8: ____ /12
Problem 9: ____ /4

For the final project, how many hours did you spend working on the project for THIS class (combining projects for this and other classes like Neuromechanics or Capstone, is great, but I just want to know how many hours you counted towards this class): 24 hours

Please evaluate your team members for the final project as well. IF they all participated equally, you can leave this blank. If they did not, please list their name(s) and give them a score out of 10 based on how well they participated with your team: I was the only team member

Problem 1 (1 pts each) -

a) T F We use the Jacobian Pseudo-inverse for IK algorithms as opposed to the Jacobian Transpose method because it is robust to singularities in the Jacobian.

b) T F The Recursive Newton-Euler formulation for dynamics gives a different set of joint torques (given q, $\dot{q}$, $\ddot{q}$) than the Euler-Lagrange formluation.

c) T F Generalized coordinates (q) for the Lagrange-Euler formulation are composed of a vector of the joint positions of a robot.

d) T F The DH parameterization is the only kinematic parameterization that will work to model the kinematics and dynamics for serial link robots.

e) T F Unique analytical inverse kinematic solutions to reach a Cartesian pose for a seven degree of freedom robot arm exist, but are often too difficult to find which is why we used numerical methods.

f) T F If we were to model link flexibility that could be expressed as a function of our generalized coordinates for a robot, the Euler-Lagrange formulation would still work to derive the equations of motion and would be grouped with gravity in the potential energy terms.

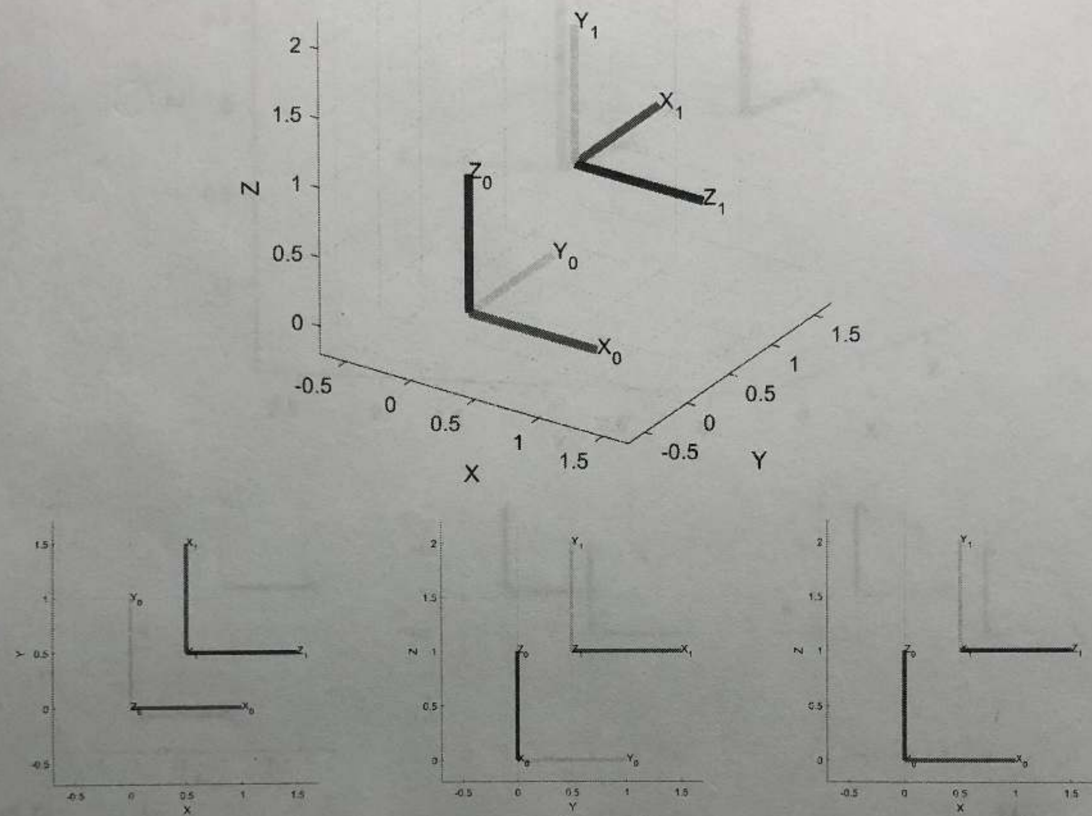
g) T F Rotational inertia for a robot link rotating about its joint axis is always more than rotational inertia for that same link rotating about its center of mass.

h) T F The following are ALL characteristics of a rotation matrix that is part of the SO(3) group: a)it must have orthonormal columns and rows, b)its determinant must be 1, c)its inverse is the same as its transpose

Problem 2 -

For all of the following, find the requested homogenous transform, obtained by inspection and basic matrix multiplication (if necessary):

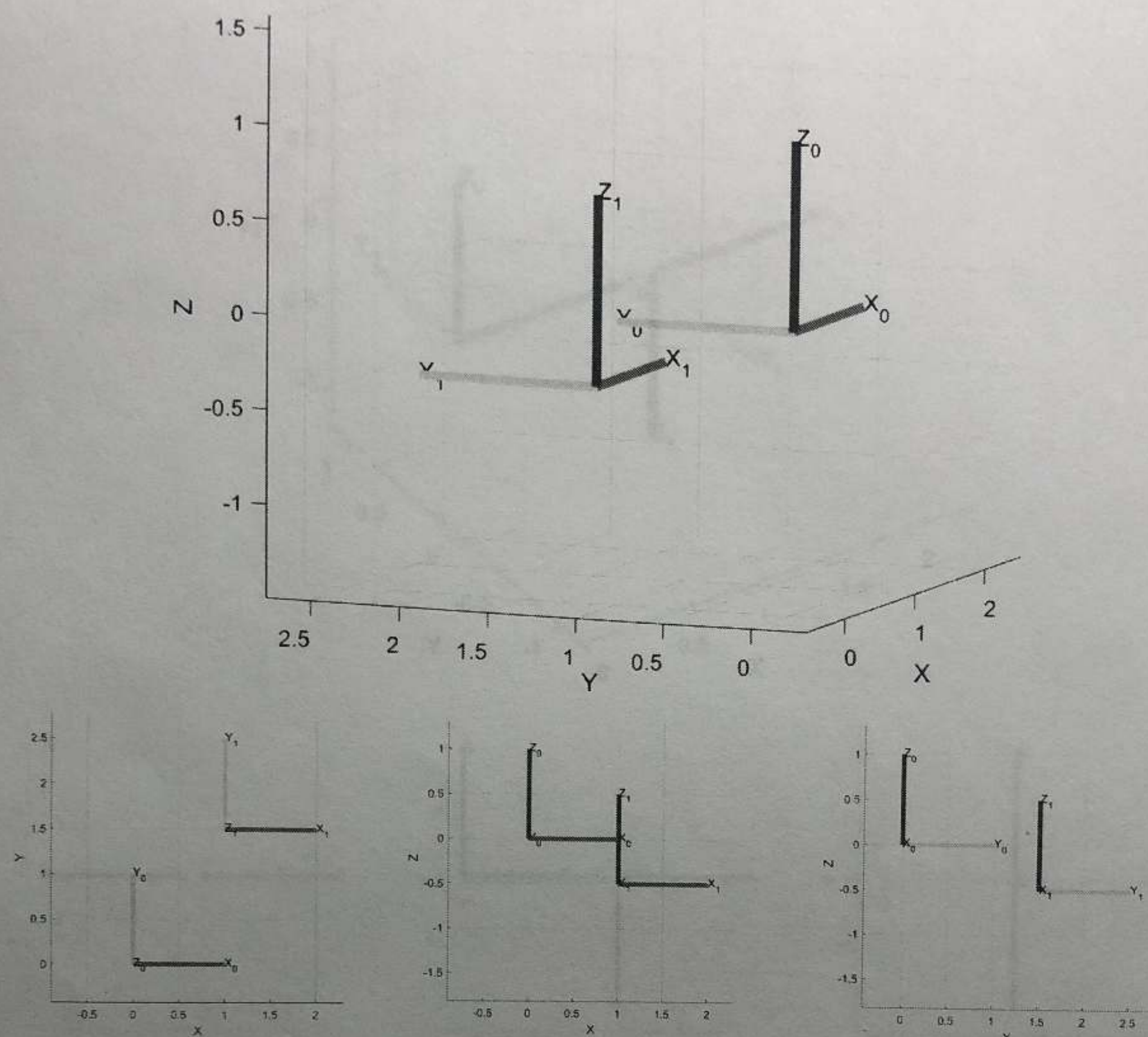
a) (3 pts) Given:



Find T_1^0: $R_z(\frac{\pi}{2})\, R_x(\frac{\pi}{2})\, t_x(0.5)\, t_y(0.5)\, t_z(1.0)$

$$\begin{bmatrix} 0 & 0 & 1 & 0.5 \\ 1 & 0 & 0 & 0.5 \\ 0 & 1 & 0 & 1.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

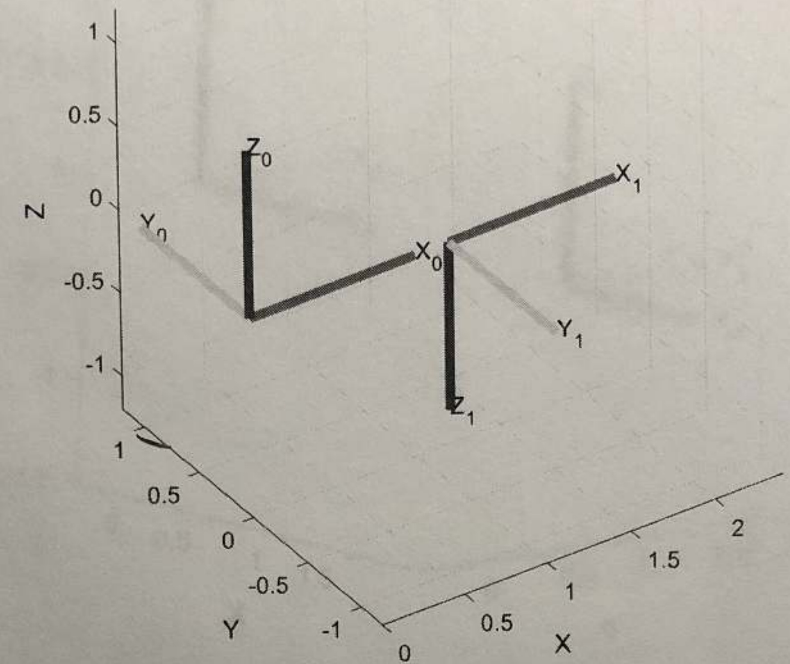
b) (3 pts) Given:

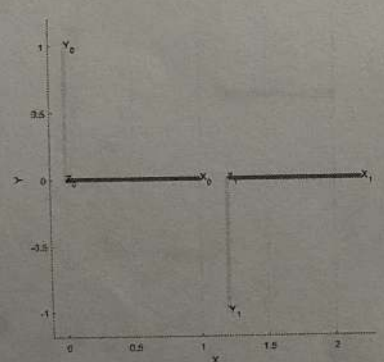
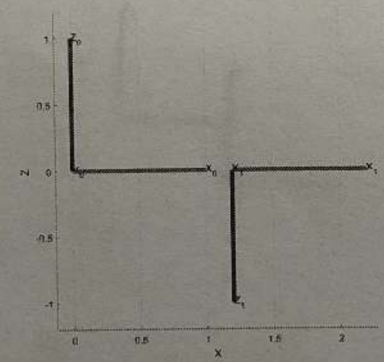
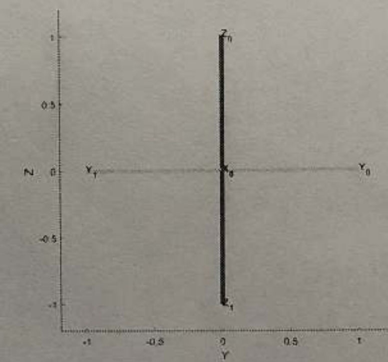


Find T_0^1:

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1.5 \\ 0 & 0 & 1 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

c) **(3 pts) Given:**

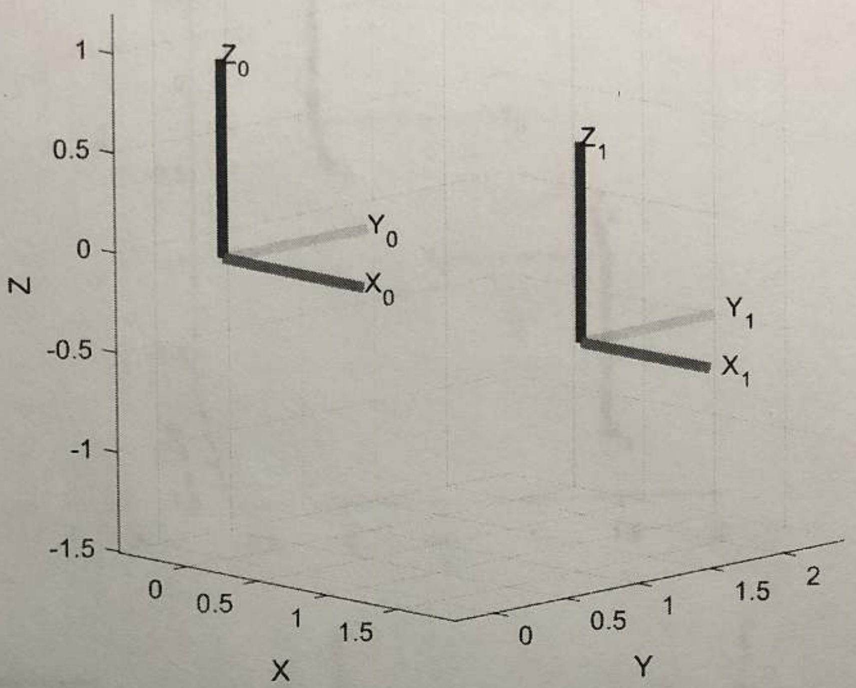


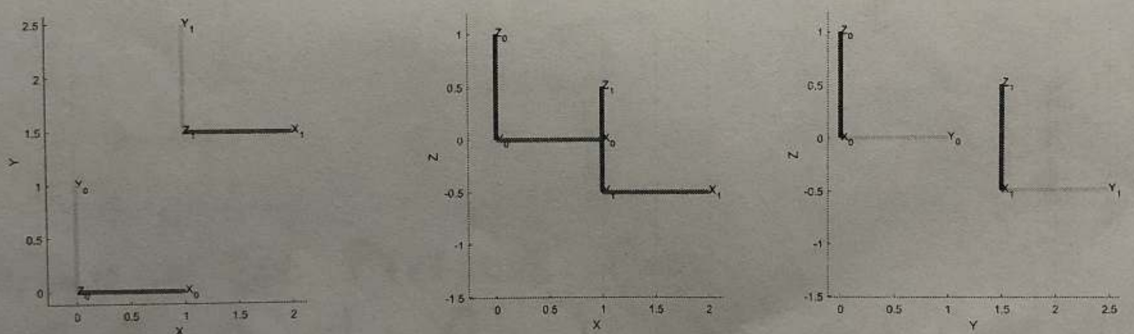




Find T_1^0:

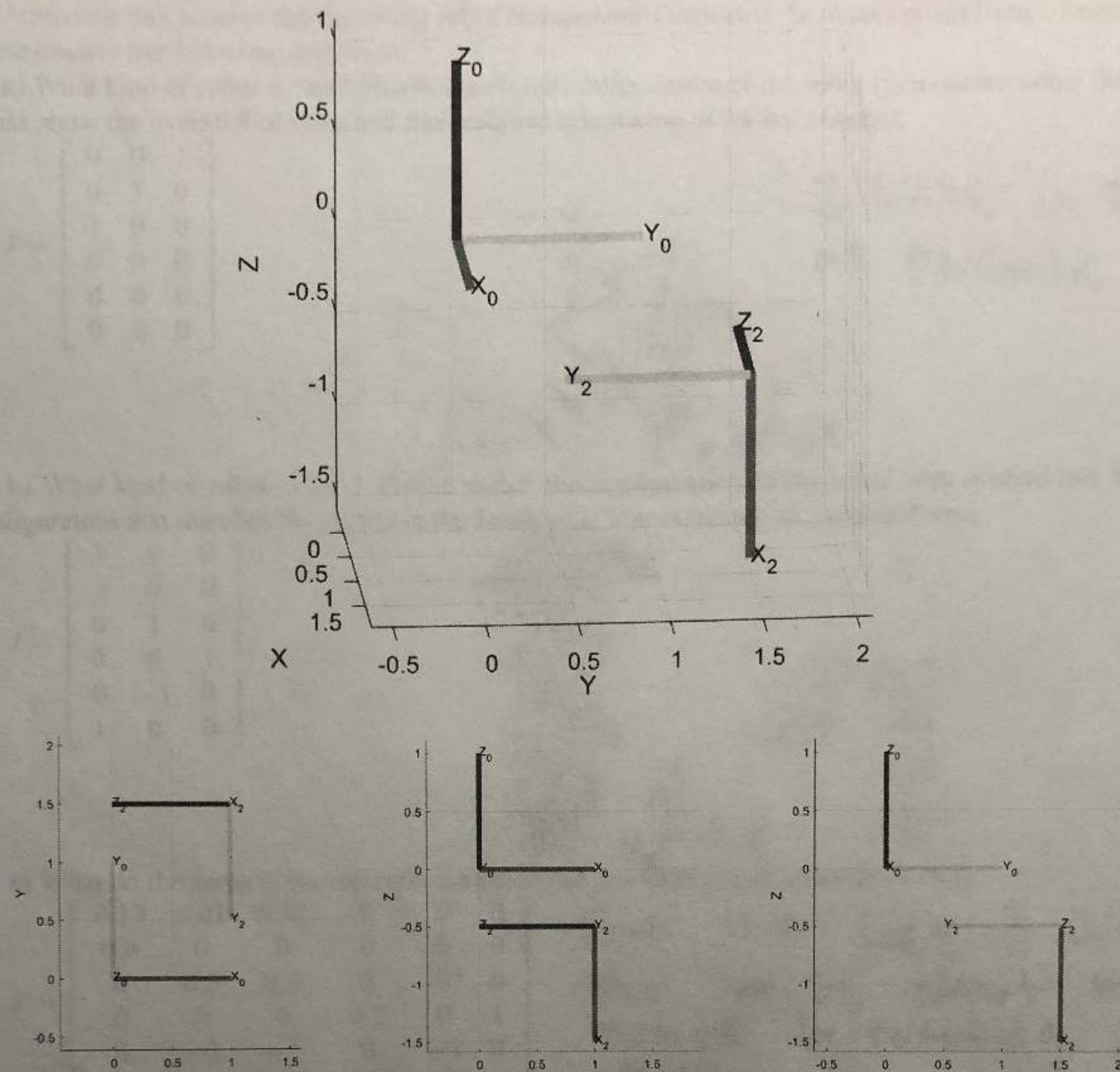
$$\begin{bmatrix} 1 & 0 & 0 & 1.25 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

d) (6 pts) Given:





AND T_1^0:
$$\begin{bmatrix} 1 & 0 & 0 & 1.0 \\ 0 & 1 & 0 & 1.5 \\ 0 & 0 & 1 & -0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Find T_2^1 and T_1^0: → See Previous Page

b.

T_2:
$$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & -0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

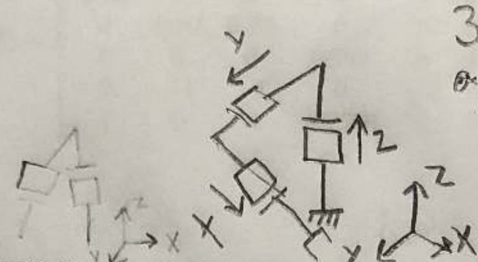
Problem 3 (4 pts each) -

Assuming that none of the following robot manipulator Geometric Jacobians result from a singular configuration, please answer the following questions:

a) What kind of robot is this? Please sketch the configuration of the robot (you cannot know the link lengths, but please show the overall # of links and their relative orientation to the base frame).

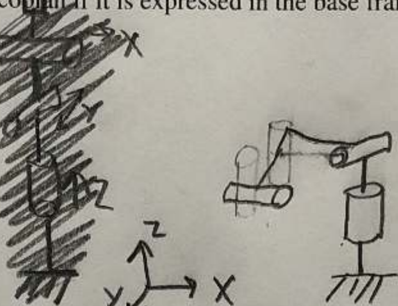
$$J = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3 Links, in Z, then Y, then X
all Prismatic Joints



b) What kind of robot is this? Please sketch the configuration of the robot with relative link lengths and in the configuration that matches the values in the Jacobian if it is expressed in the base frame.

$$J = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$



c) What do the zeros in the top right corner of this Jacobian tell us about this robot?

$$J = \begin{bmatrix} 0.15 & 0.01 & 0.32 & 0 & 0 & 0 \\ 0.6 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.6 & 0.3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.7 & 0 & 1 \\ 0 & -1 & -1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -0.7 & 0 & 0 \end{bmatrix}$$

That the Last 3 Joints most Likely ~~the~~ wrist, cannot make any translation motion by rotating, Likely the wrist is in a singularity or have no Length, which is why they cant move the end effector Linearly

d) For the same Jacobian as in problem c), if we assume that this Jacobian was found for the end effector, what do the entries in the first three columns tell us about how the robot can move (be specific)?

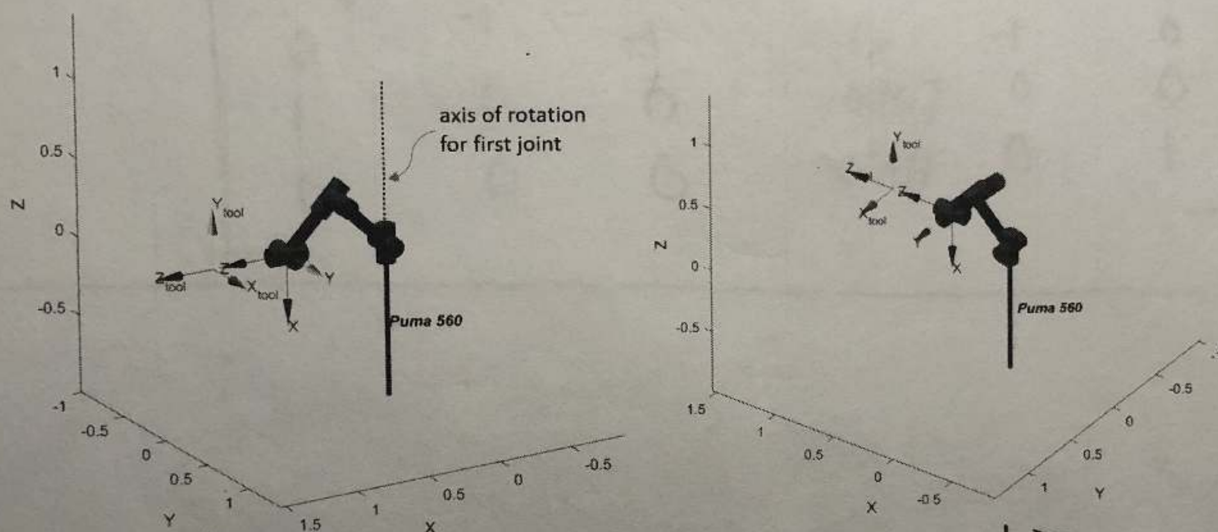
- The first Joint can move the robot EE in X a Little, and a Lot in Y by rotating around Z.
- The Second Joint can move the robot EE in X only a tiny ammount but mostly in Z by rotating about Y.
- the third Joint can move the robot end effector in X and Z almost equally by rotating around Y.

with these combonations the robot end effector can move in any direction except rotate in X with the first 3 Joints.

Problem 4 (12 pts) -

For a puma robot, given the Jacobian listed below and calculated <u>at</u> the end effector, but expressed in the base frame (see labeled axes on plots for base frame orientation), please shift this Jacobian to the tool tip (0.5 meters in the z-direction of the end effector frame) shown in the figure and expressed in the tool tip frame shown (i.e. find J^{tool}_{tool}).

$$J^0_{ee} = \begin{bmatrix} 0.15 & 0.01 & 0.32 & 0 & 0 & 0 \\ 0.60 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.60 & 0.29 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.707 & 0 & 1 \\ 0 & -1 & -1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -0.707 & 0 & 0 \end{bmatrix}$$



$$J^n_n(q) = \begin{bmatrix} R^n_0 & 0 \\ 0 & R^n_0 \end{bmatrix} J^0_n(q).$$

$$R^{Base}_{EE} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$J^{EE}_{EE}(q) = \begin{bmatrix} R^{EE}_{base} & 0 \\ 0 & R^{EE}_{base} \end{bmatrix} J^0_{ee}$$

$$R^{EE}_{Base} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$J^{EE}_{EE} = \begin{bmatrix} 0 & -0.6 & -0.29 & 0 & 0 & 0 \\ 0.6 & 0 & 0 & 0 & 0 & 0 \\ 0.15 & 0.01 & 0.32 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0.707 & 0 & 0 \\ 0 & -1 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0.707 & 0 & 1 \end{bmatrix}$$

See next Page

$$J^{Tool}_{Tool}(q) = \begin{bmatrix} R^{Tool}_{EE} & 0 \\ 0 & R^{Tool}_{EE} \end{bmatrix} J^{EE}_{EE}$$

$$R^{EE}_{Tool} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad R^{Tool}_{EE} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\boxed{J^{Tool}_{Tool}(q) = \begin{bmatrix} 0.6 & 0 & 0 & 0 & 0 & 0 \\ 0.0 & 0.6 & 0.29 & 0 & 0 & 0 \\ 0.15 & 0.01 & 0.32 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -0.707 & 0 & 0 \\ 0 & 0 & 0 & 0.707 & 0 & 1 \end{bmatrix}}$$

Problem 5 (9 pts) -

The following form of the dynamics are for a serial-link manipulator

$$M(q)\ddot{q}+C(q,\dot{q})\dot{q}+G(q)+F(\dot{q})=\tau$$

In order to do numerial integration (or simulation) for a robot, we must put the equation above in state variable from (which is $\dot{x}=f(x,u)$). Please define the state variable x, the input vector u, and write the equations for a robot arm in state variable form.

$$X=\begin{bmatrix}\dot{\theta} \\ \theta\end{bmatrix} \qquad U=\begin{bmatrix}\tau_1 \\ \tau_2 \\ \cdots \\ \tau_n\end{bmatrix} \text{ for n Links}$$

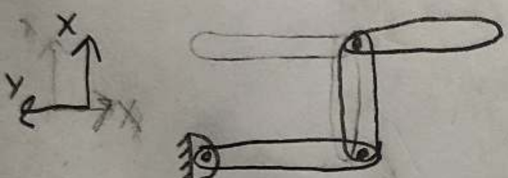
$$\dot{X}=\begin{bmatrix} M(q)^{-1}\left(U-C(q,\dot{q})\dot{q}-G(q)\right) \\ \dot{\theta}\end{bmatrix}$$

$$X_{t+1}=X_t+\Delta t\,\dot{X}_t$$

Problem 6 (12 pts each) -

a) For a 3 degree of freedom RRR robot, and the information below, please find the instantaneous (because you are given the current Jacobian instead of the Jacobian as a function of q) kinetic energy of the last link in terms of the variables given and $\dot{q}$. Show your work and any intermediate steps to make it clear that you understand the process, and then explain explicitly how this relates to finding the total generalized mass matrix ($M(q)$) of the robot for the current time or configuration of the robot.

$$J_{COM,3} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad m_3 = 2\,kg, R_3^0 = \begin{bmatrix} 0.707 & -0.707 & 0 \\ 0.707 & 0.707 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad I_{3,COM} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$K = \frac{1}{2}\dot{q}^T\left[\sum_{i=3}^{3} m_i J^T_{v_i com} J_{v_i,com} + J^T_{w_i} R^0_i I_i R^i_0 J_{w_i}\right]\dot{q}$$

$$J_{v_i,com} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$J_{w_i,com} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$K = \frac{1}{2}\dot{q}^T\left[\begin{bmatrix} 2 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}\right]\dot{q}$$

$$\boxed{K = \frac{1}{2}\dot{q}^T\begin{bmatrix} 3 & 1 & 3 \\ 1 & 3 & 1 \\ 3 & 1 & 3 \end{bmatrix}\dot{q}}$$

This kenetic energy equation can be used to identify the mass matrix from the equation $K = \frac{1}{2}mv^2$. in our vairiant $\frac{1}{2}$, and v^2 are included in the $\dot{q}^T$, $\dot{q}$ and $\frac{1}{2}$ constant so' the mass matrix for this configuration is

$$\begin{bmatrix} 3 & 1 & 3 \\ 1 & 3 & 1 \\ 3 & 1 & 3 \end{bmatrix}$$

at the third joint. if all othe Joints were summed then you could find the total mass matrix in this configuration.

Problem 7 (12 pts) -

Given the information below calculate a single step of the forward recursion (calculating α, ω, and acceleration at the center of mass and end of the link) of the Recursive Newton-Euler method.

$$a_{e,0} = a_{c,0} = \omega_0 = \alpha_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad R_1^0 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad r_{0,c_1}^1 = \begin{bmatrix} 0.2 \\ 0 \\ 0 \end{bmatrix}, \quad r_{0,1}^1 = \begin{bmatrix} 0.4 \\ 0 \\ 0 \end{bmatrix}$$

$$Z_0^0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$q = \begin{bmatrix} \frac{\pi}{2} \\ -\frac{\pi}{4} \\ \frac{\pi}{4} \end{bmatrix}, \quad \dot{q} = \begin{bmatrix} -\frac{\pi}{6} \\ -\frac{\pi}{6} \\ \frac{\pi}{3} \end{bmatrix}, \quad \ddot{q} = \begin{bmatrix} \frac{\pi}{6} \\ 0 \\ \frac{\pi}{6} \end{bmatrix} \quad R_1^0 = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$w_1^1 = R_1^0 w_0^0 + R_0^1 Z_0^0 \dot{q}_1$$

$$\alpha_1^1 = R_1^0 \alpha_0^0 + R_0^1 Z_0^0 \ddot{q}_1 + w_1^1 \times R_0^1 Z_0^0 \dot{q}_1$$

$$a_{com,1}^1 = R_1^0 a_{e,0}^0 + \alpha_1^1 \times R_{0,com,1}^1 + w_1^1 \times (w_1^1 \times r_{0,com1}^1)$$

$$a_{e,1}^1 = R_1^0 a_{e,0}^0 + \alpha_1^1 \times R_0^1 + w_1^1 \times (w_1^1 \times r_0^1)$$

$$w_1^1 = [0, 0, 0] + [0, 0, -0.524] = [0, 0, -0.524]$$

$$\alpha_1^1 = [0, 0, 0] + [0, 0, 0.524] + [0, 0, 0] = [0, 0, 0.5$$

$$a_{com,1}^1 = [0, 0, 0] + [0, 0, 0.524] + [-0.055, 0, 0] = [-0.055, 0, 0.$$

$$a_{e,1}^1 = [0, 0, 0] + [0, 0, 0.524] + [-0.110, 0, 0] = [-0.110, 0, 0.524$$

computed in python

$$\boxed{\begin{aligned} w_1^1 &= [0, 0, -0.524] \\ \alpha_1^1 &= [0, 0, 0.524] \\ a_{com,1}^1 &= [-0.055, 0, 0.524] \\ a_{e,1}^1 &= [-0.110, 0, 0.524] \end{aligned}}$$

Problem 8 (12 pts) -

Given the information below calculate a single step of the backward recursion (calculating f, τ) of the Recursive Newton-Euler method. Note that we are starting at the last link of a robot, but that there is an external force (F_{ext}) as listed below. You should report all of the reaction wrench (forces and torques), but make it clear which torque would correspond with the joint motor.

$$F_{ext}^{ext} = \begin{bmatrix} 50 \\ 100 \\ 50 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad R_3^0 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_{ext}^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_3^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.707 & -0.707 \\ 0 & 0.707 & 0.707 \end{bmatrix}, \quad m_3 = 1.2kg$$

$$a_{e,2}^2 = \begin{bmatrix} -0.5 \\ 0.2 \\ 0 \end{bmatrix}, \quad a_{c,3}^3 = \begin{bmatrix} -0.2 \\ 0.8 \\ 0 \end{bmatrix}, \quad a_{e,3}^3 = \begin{bmatrix} -0.272 \\ 1.09 \\ 0 \end{bmatrix} \quad \dot{q} = \begin{bmatrix} -\frac{\pi}{6} \\ -\frac{\pi}{6} \\ \frac{\pi}{3} \end{bmatrix} \quad g = \begin{bmatrix} 0 \\ -9.81 \\ 0 \end{bmatrix}$$

$$I_{link,3} = \begin{bmatrix} 0.01 & 0 & 0 \\ 0 & 0.01 & 0 \\ 0 & 0 & 0.01 \end{bmatrix}, \quad r_{2,c_3}^3 = \begin{bmatrix} 0.2 \\ 0 \\ 0 \end{bmatrix}, \quad r_{3,c_3}^3 = \begin{bmatrix} -0.2 \\ 0 \\ 0 \end{bmatrix}, \quad \alpha_3^3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \omega_3^3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$F_3^3 = R_{ext}^3 F_{ext}^{ext} - m_3 (R_0^3 g) + m_3 a_{c,3}^3$$

$$\tau_3^3 = R_{ext}^3 \tau_{ext}^{ext} - F_3^3 \times r_{2,c3}^3 + (R_{ext}^3 F_{ext}^{ext}) \times r_{3,c3}^3 + \dots$$
$$I_{3,com}^3 \alpha_3^3 + w_3^3 \times (I_{3,com}^3 \times w_3^3)$$

$$\boxed{F_3^3 = \begin{bmatrix} 37.988 \\ 100.96 \\ 50 \end{bmatrix} \qquad \tau_3^3 = \begin{bmatrix} 0.01 \\ -19.99 \\ 46.010 \end{bmatrix} \qquad \tau \text{ at motor} = 40.010}$$

Computed in Python

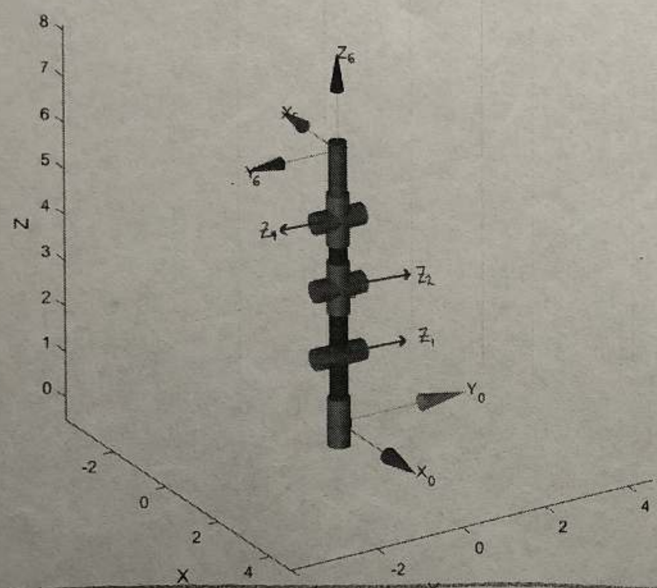
$$\tau_3^3 = [0,0,0] + [6, 10, -20] + [0, -10, 20] + [0.01, 0.01, 0.01] + [0,0,0] + [0.01]$$

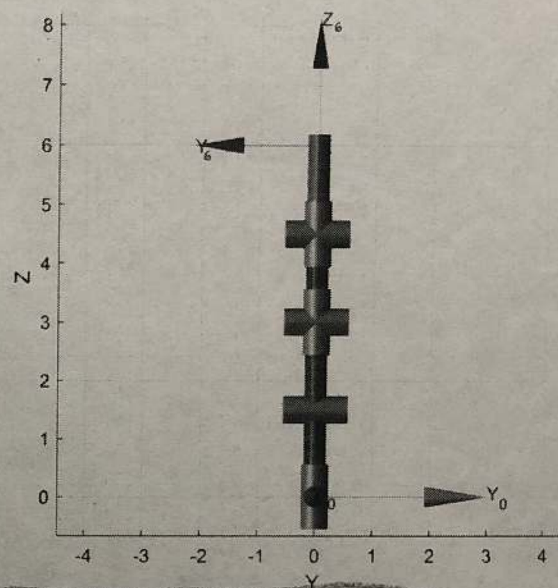
$$F_3^3 = [50, 100, 50] + [11.772, 0, 0] + [-0.24, 0.96, 0]$$

Problem 9 (4 pts) -

If they exist, find at least two different wrenches that when applied to the end-effector for the robot shown below would cause no motion or in other words, no joint torque. IF they exist, report the 6x1 wrench vector in the base frame, otherwise, say if it doesn't exist. The Jacobian for the problem below is as follows:

$$J_6^0(q) = \begin{bmatrix} 0 & 4.5 & 3 & 0 & -1.5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$





wrench 1 = $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

wrench 2 = $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

Check with $J^T W = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

which is true

for both wrenches